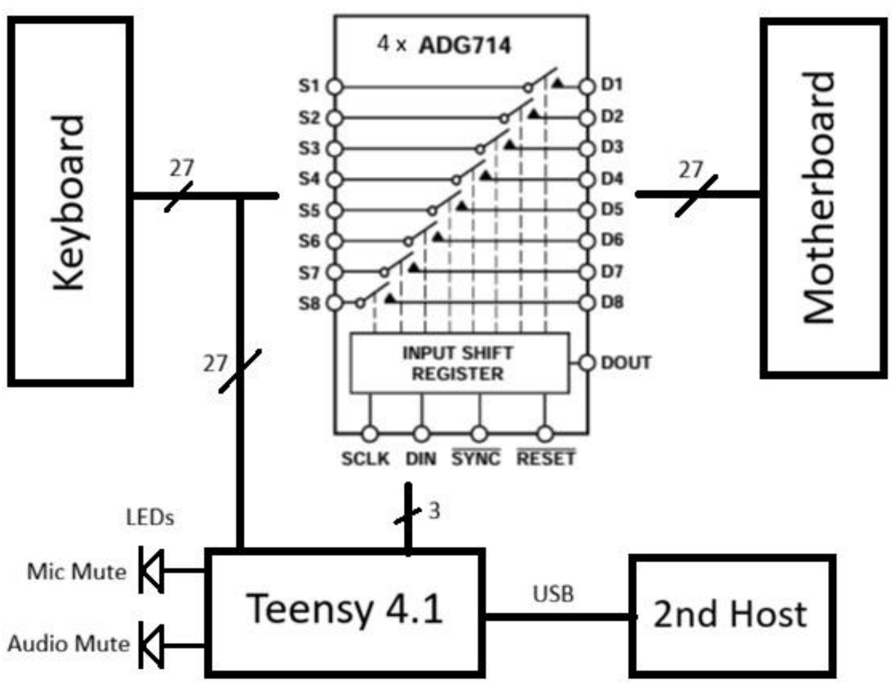
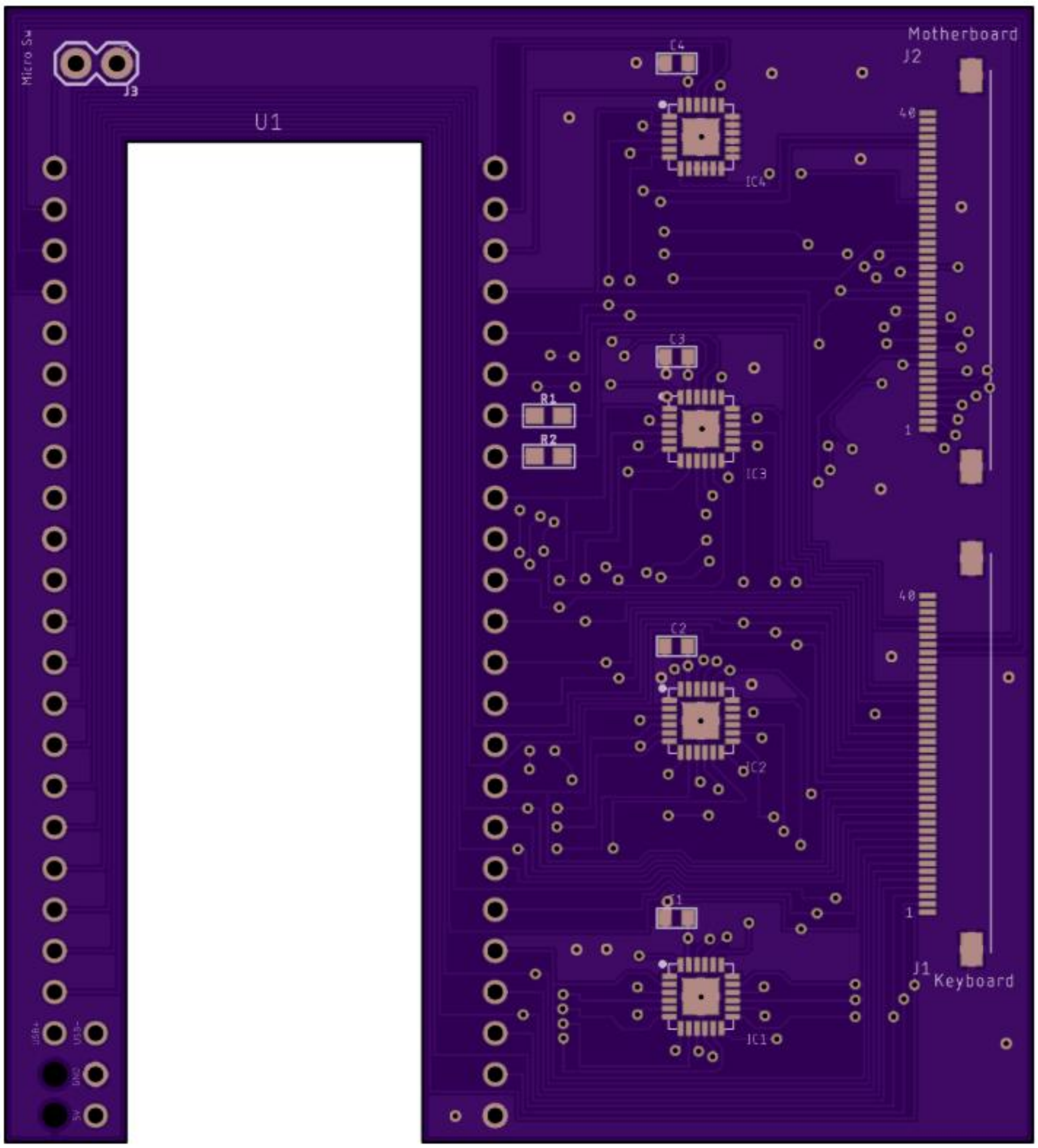
Lenovo ThinkPad P15 Gen 1 Keyboard – Dual Connection

In this project, a Lenovo Thinkpad P15 Gen 1 laptop keyboard shown below is either controlled by a Teensy 4.1 that is cabled via USB to a host computer, or the row and column pins are connected to the motherboard (MB) as they would be in a normal laptop. The backlight is not used and the trackpoint is controlled by a separate circuit.

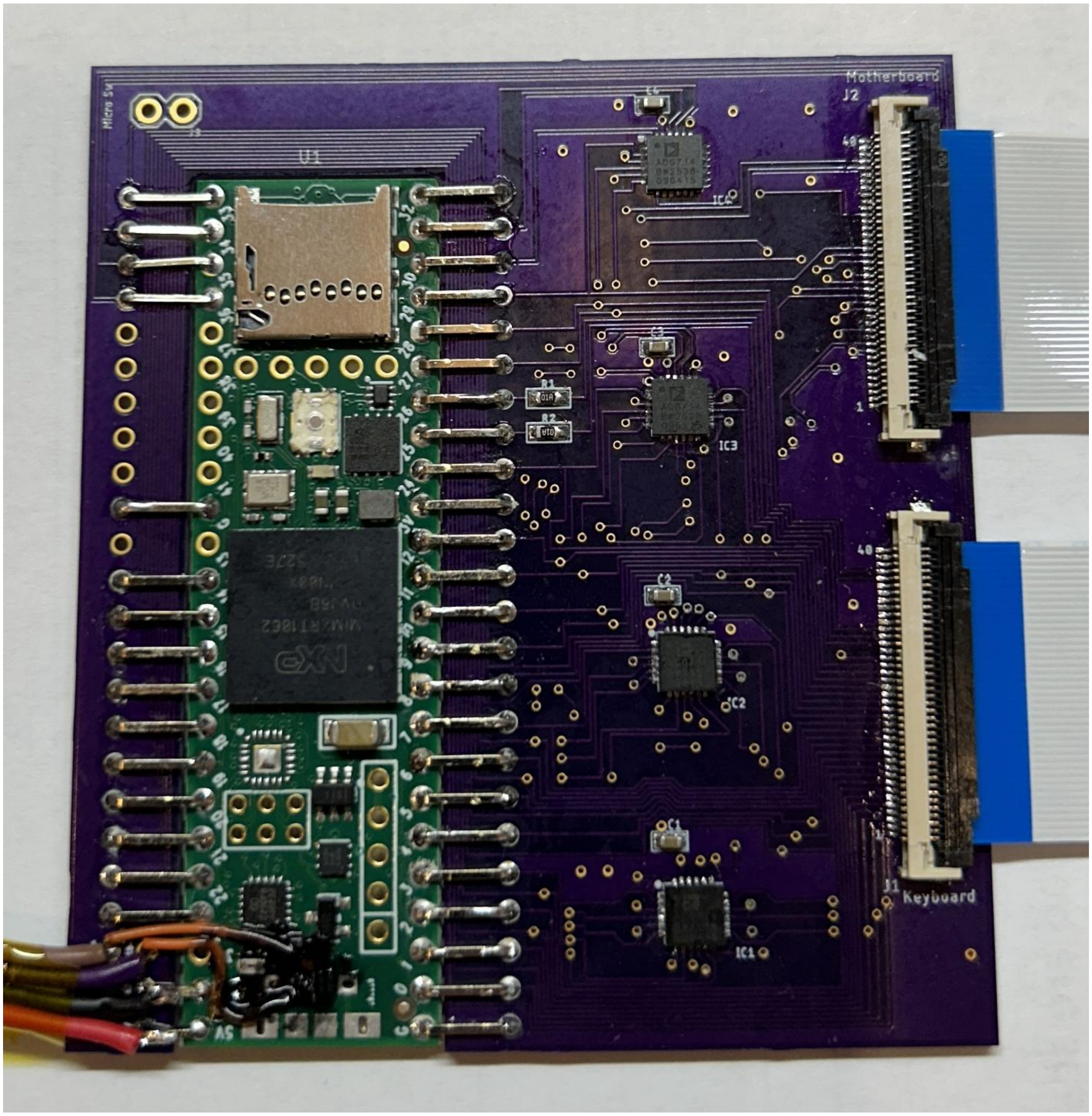
Four ADG714 octal switches connect the 27 keyboard row & column signals to the MB (see below). Holding the Fn key down for more than 2 seconds signals to the Teensy to switch control to the opposite host. The Teensy shifts in all 0's or all 1's to open or close the switches. The Audio Mute LED is illuminated when the MB is in control of the keyboard and the Mic Mute LED shows the Teensy is in control. The Teensy is powered by the USB 5 volts and wakes up with the switches turned off.



The circuit board shown below was designed in Eagle and fabricated by OSH Park. The Teensy 4.1 sits in the cutout and is wired to the board with low profile jumpers. Closing the ADG714 switches will connect the keyboard to the MB. The Teensy is still connected to the keyboard but the code configures all 27 Teensy GPIO's as inputs so they don't interfere with the MB scan. The Fn "Hotkey" is unique in Lenovo keyboards because it is not part of the row/column matrix. One side of the Fn key switch is always tied to ground and the other side is always available to be read by the Teensy and the MB. This allows the Teensy to always be able to detect when the Fn key is held down. Pads for reading an optional microswitch can be seen in the upper left corner.



The assembled circuit board with all components is shown below



The USB connector and push button switch are the tallest components on the Teensy. A hot air rework station was used to remove the connector. The rubber and plastic on the push button switch melted from the hot air so they were cut away with wire cutters. Equal length 30-gauge wire wrap wire was soldered to the USB D+ and D- surface mount pads previously used by the USB connector. These wires go to pads on the lower left of the circuit board and are coated with liquid electrical tape for protection.

The USB cable is made from individual wires cut to the same length (20cm) and wrapped with Kapton tape. The USB A connector soldered to the wires does not have a plastic cover in order to reduce its height (see below).

This end view picture shows the low profile of the assembled board which allows it to fit inside the laptop case.

The keyboard matrix is given below. The 40 pin FPC pin number is listed first, followed by the connection to the Teensy GPIO numbers and the Lenovo schematic signal name.

P15 Keyboard Matrix

FPC pins – GPIO # Signal Name	1-23 SENSE3	2-0 SENSE7	3-22 SENSE6	5-21 SENSE4	6-2 SENSE1	8-3 SENSE2	9-19 SENSE0	12-5 SENSE5
4-1 DRV14	L-SHIFT		R-SHIFT					
7-20 DRV0	TAB		Z	A	1	Q	TILDE	ESC
10-4 DRV4	Y	N	M	J	7	U	6	H
11-18 DRV2	F3		C	D	3	E	F2	F4
13-17 DRV1	CAPS		X	S	2	W	F1	
14-6 DRV3	T	B	V	F	4	R	5	G
15-16 DRV6	F7		PERIOD	L	9	O	F8	
16-7 DRV7	[	SLASH		;	0	P	MINUS	QUOTE
17-15 DRV5	]		COMMA	K	8	i	=	F6
18-8 DRV15			R-CNTL				L-CNTL	
19-14 DRV13		R-ALT			PNTSCN			L-ALT
20-9 DRV9	GUI	RIGHT			F12	KP .	INS	
21-10 DRV12	KP =	LEFT			END	KP [	HOME	UP
22-11 DRV10		DOWN			F11		DEL	
23-12 DRV8	BKSPC	SPACE	ENTER	\	F10		F9	F5
24-24 DRV11	KP]				PGDN	KP BK	PGUP	
36-25 DRV16	KP /	KP +	KP 9	KP 7	KP 8	KP *	KP -	NUMLK
37-26 DRV17	KP 5	KP 0	KP ENT	KP 2	KP 3	KP 6	KP 1	KP 4

The trackpoint left, middle, and right buttons are part of the keyboard and are scanned by the Teensy. The Teensy sends the button status over USB along with the keyboard key presses. The Teensy is programmed to put pull up resistors on each of these button pins so that a button press is seen as a logic 0.

Trackpoint Middle Button = GPIO 34 on FPC pin 35

Trackpoint Right Button = GPIO 35 on FPC pin 34

Trackpoint Left Button = GPIO 36 on FPC pin 33

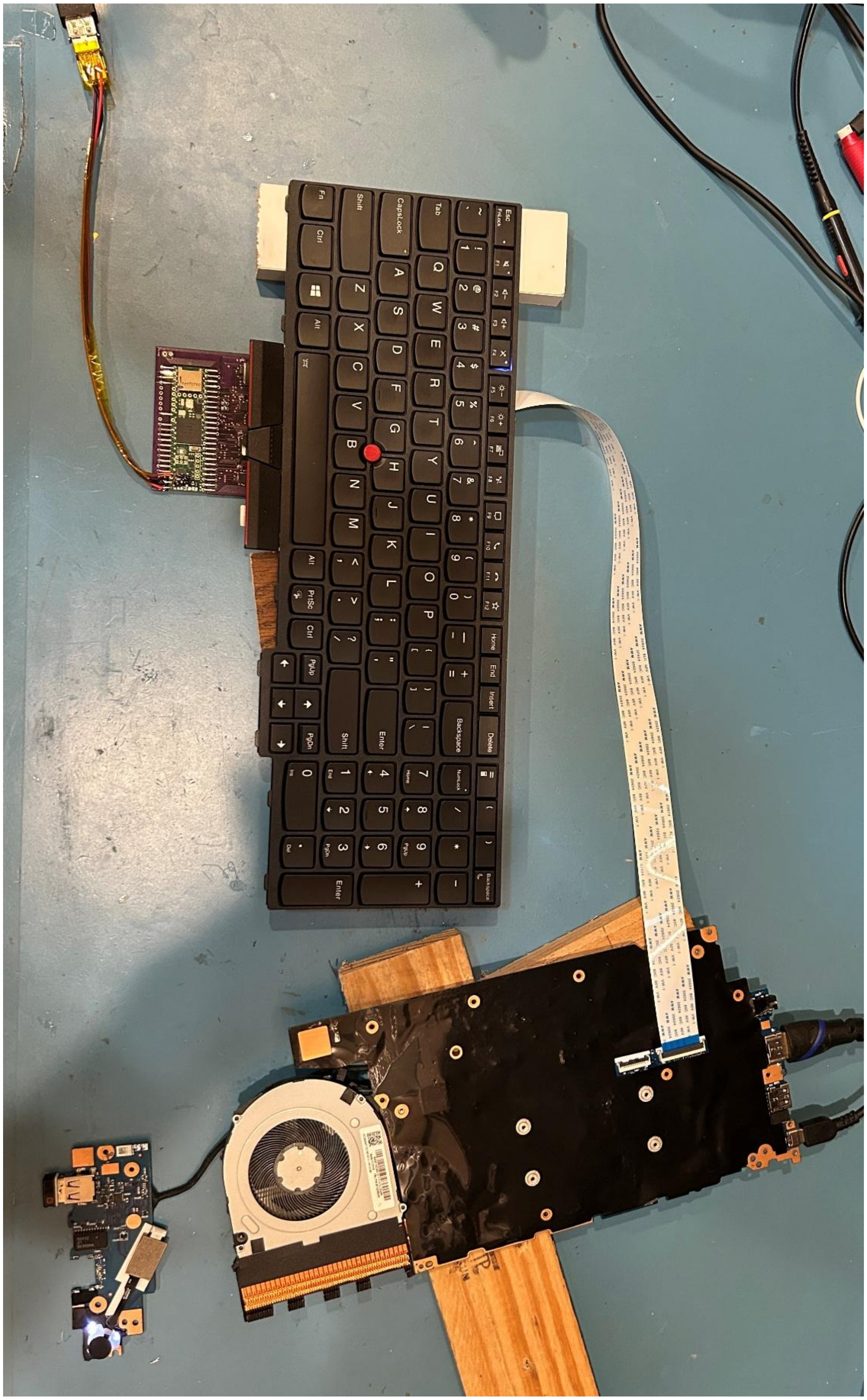
The return for all of the Trackpoint buttons is on FPC pin 32 and is tied to ground

The Fn (Hotkey) on FPC pin 29 is not part of the switch matrix and is connected to Teensy GPIO 27 (programmed to have a pullup resistor). The Fn return on FPC pin 30 is tied to ground.

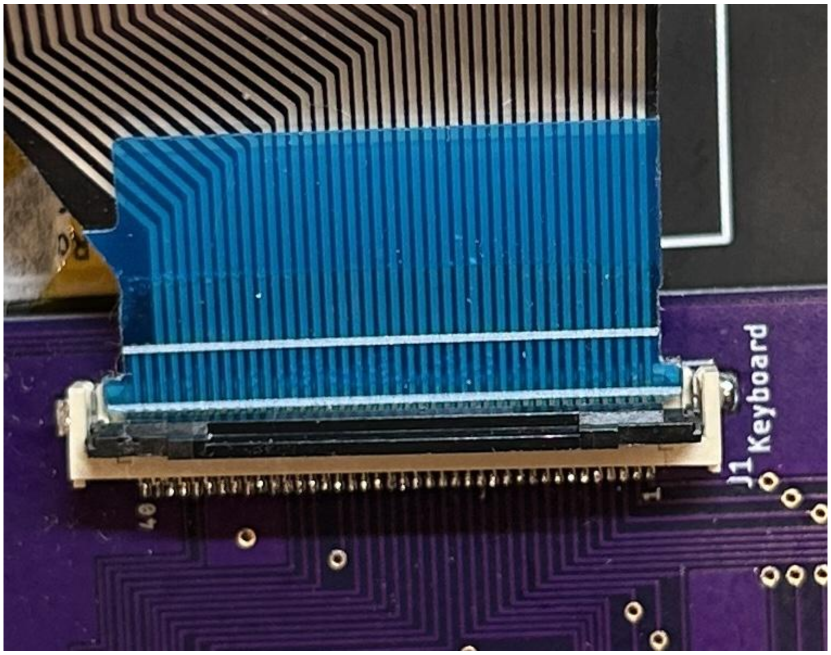
The Mic Mute LED cathode on FPC pin 28 is routed through a 100 ohm resistor to Teensy GPIO 26. The Speaker Mute LED cathode on FPC pin 27 is routed through a 100 ohm resistor to Teensy GPIO 25. The Teensy will take these GPIO pins to ground to turn on the LEDs. The anodes of both LEDs are connected to 3.3 volts on FPC pin 25.

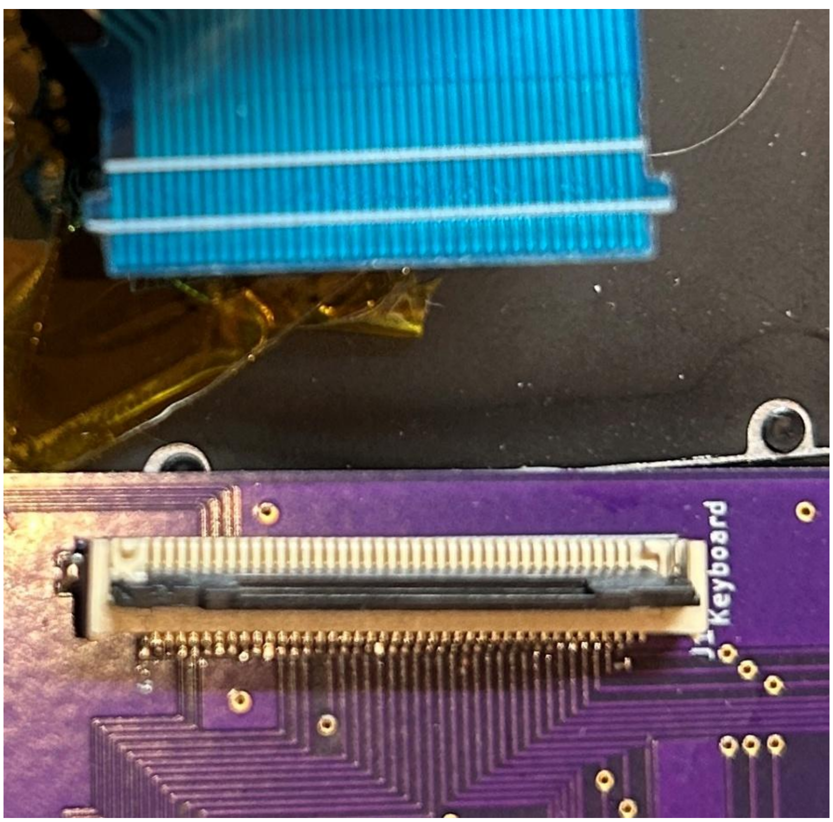
Teensy GPIO 33 is routed to the corner of the circuit board for an optional micro switch.

The keyboard, Teensy switch board and motherboard are shown below.

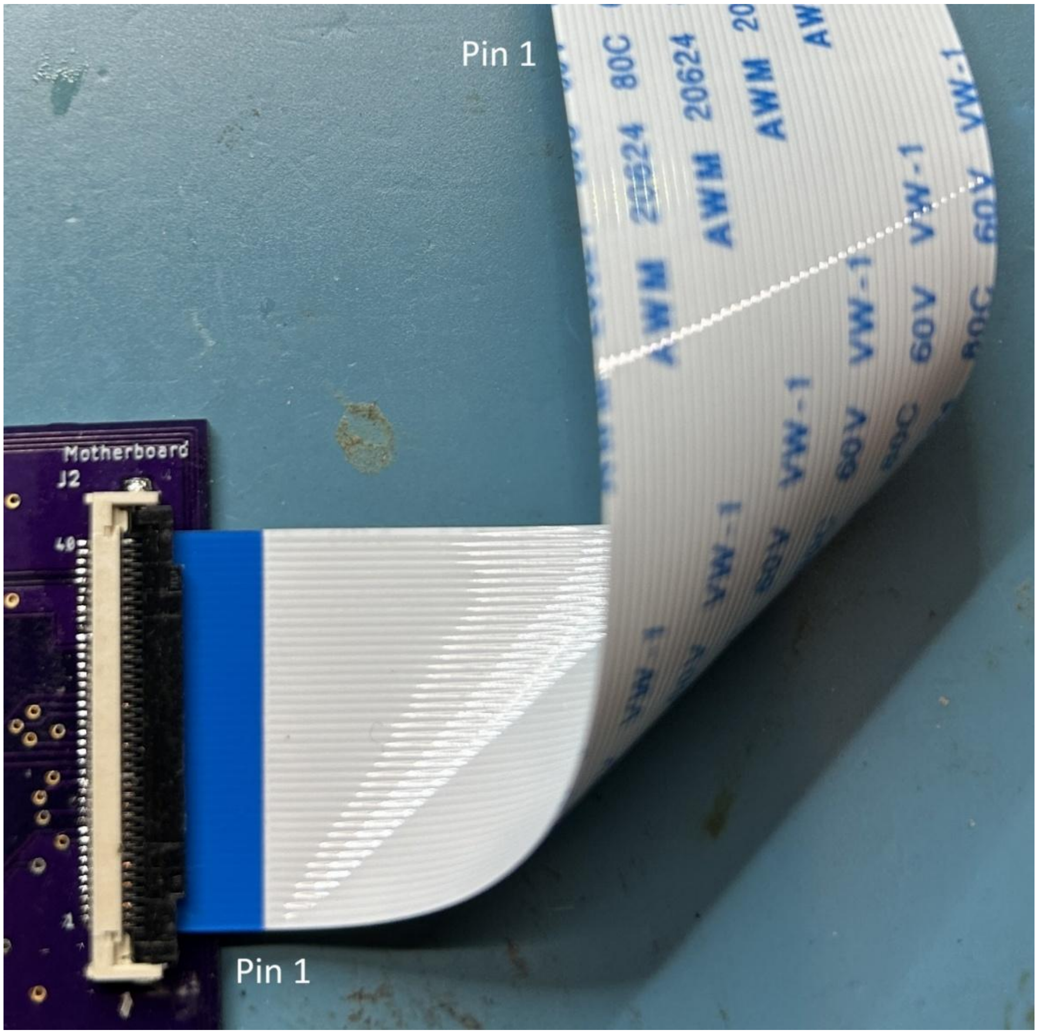


Both 40 pin 0.5mm pitch FPC connectors on the Teensy board will accept cables with standard size ear tabs but the keyboard cable has extra wide ears that do not fit in the notch (see below). Testing shows the connector is still able to lock the cable down and make good connection so I didn't have to cut off the ears. I was unable to find long FPC cables with ears that would go from the Teensy board to the motherboard. Standard FPC cables without ears are used and they lock down without problems.





All of the FPC connectors have bottom contacts. The 50cm long FPC cable from the Teensy board to the MB is known as a Type B (contacts are on the bottom side on one end and on the top side on the other end). This FPC cable must be folded 90 degrees which puts the contacts on the bottom side at both ends and keeps pin 1 at the correct location (see below). Note that there is no industry standard for the pin 1 location and the popular AliExpress FPC breakout board has pin 1 labeled on the other end.



If there had been no need to fold the cable to make everything fit in the laptop, a Type A FPC cable would have been used. Also, the Teensy board traces would have been designed to swap the signals end for end on the MB FPC connector (the equivalent of folding the FPC cable).

Initial testing showed some of the keys were intermittent when the switchers were connected to the MB. One of the ADG714 chips (IC2) was always the one with the bad switches. To make sure the problem was fixed, I lowered the serial clock speed when loading all of the ADG714's from 1MHz to 1KHz and the code always loads the byte twice just to be sure. The solder connections were reheated to IC2 and a ground jumper was added (see below) to supplement the less-than-ideal ground trace in the board.